1 Parsing Native Exports

1.1 Learning objectives

After completing this chapter, you will be able to:

- Parse BibTeX files using `bib2df` and `RefManageR`
- Import Web of Science plain-text and Scopus CSV exports via `bibliometrix`
- Standardize field names across formats into a unified schema
- Diagnose common parsing failures and missing fields

1.2 Setup

```
library(tidyverse)
library(bib2df)
library(RefManageR)
library(bibliometrix)
library(glue)

set.seed(20260509)

source(here::here("R", "sci_palette.R"))
```

1.3 Conceptual background

Not all bibliometric data comes from an API. Researchers frequently begin with files exported from database search interfaces: BibTeX (`.bib`) from Google Scholar or Zotero, RIS (`.ris`) from Scopus or PubMed, plain text from Web of Science, or CSV from Scopus. Each format encodes the same information — authors, title, year, journal, DOI — in different structures.

The `bibliometrix` package [@aria2017] provides `convert2df()`, a versatile function that reads multiple export formats and returns a standardized data frame. For BibTeX specifically, `bib2df` and `RefManageR` offer complementary parsing capabilities. The key challenge is not parsing per se but *standardization*: ensuring that author names, journal titles, and identifiers are represented consistently regardless of the source format.

1.4 Worked example

1.4.1 Parsing a BibTeX string

We construct a small BibTeX string inline to demonstrate parsing without external files.

```
bib_text <- '
@article{smith2023,
  author = {Smith, John A. and Doe, Jane B.},
  title = {Advances in Bibliometric Methods},
  journal = {Journal of Informetrics},
  year = {2023},
  volume = {17},
  pages = {101--115},
  doi = {10.1234/example.2023}
```

```
}
@article{chen2022,
  author = {Chen, Wei and Li, Xiao},
  title = {Citation Networks in Computer Science},
  journal = {Scientometrics},
  year = {2022},
  volume = {127},
  pages = {3201--3220},
  doi = {10.1234/example.2022}
}
'

tmp_bib <- tempfile(fileext = ".bib")
writeLines(bib_text, tmp_bib)

bib_df <- bib2df(tmp_bib)
glimpse(bib_df)
```

```
#> Rows: 2
#> Columns: 27
#> $ CATEGORY     <chr> "ARTICLE", "ARTICLE"
#> $ BIBTEXKEY    <chr> "smith2023", "chen2022"
#> $ ADDRESS      <chr> NA, NA
#> $ ANNOTE       <chr> NA, NA
#> $ AUTHOR       <list> <"Smith, John A.", "Doe, Jane B.">, <"Chen, Wei", "Li, Xi~
#> $ BOOKTITLE    <chr> NA, NA
#> $ CHAPTER      <chr> NA, NA
#> $ CROSSREF     <chr> NA, NA
#> $ EDITION      <chr> NA, NA
#> $ EDITOR       <list> NA, NA
#> $ HOWPUBLISHED <chr> NA, NA
#> $ INSTITUTION  <chr> NA, NA
#> $ JOURNAL      <chr> "Journal of Informetrics", "Scientometrics"
#> $ KEY          <chr> NA, NA
#> $ MONTH        <chr> NA, NA
#> $ NOTE         <chr> NA, NA
#> $ NUMBER       <chr> NA, NA
#> $ ORGANIZATION <chr> NA, NA
#> $ PAGES        <chr> "101--115", "3201--3220"
#> $ PUBLISHER    <chr> NA, NA
#> $ SCHOOL       <chr> NA, NA
#> $ SERIES       <chr> NA, NA
#> $ TITLE        <chr> "Advances in Bibliometric Methods", "Citation Networks in~
#> $ TYPE         <chr> NA, NA
#> $ VOLUME       <chr> "17", "127"
#> $ YEAR         <dbl> 2023, 2022
#> $ DOI          <chr> "10.1234/example.2023", "10.1234/example.2022"
```

1.4.2 Parsing with RefManageR

```r
bib_rm <- ReadBib(tmp_bib)
as.data.frame(bib_rm) |> select(title, author, year, journal, doi)
```

```
#>                                               title                          author
#> smith2023      Advances in Bibliometric Methods John A. Smith and Jane B. Doe
#> chen2022  Citation Networks in Computer Science          Wei Chen and Xiao Li
#>           year                journal                  doi
#> smith2023 2023 Journal of Informetrics 10.1234/example.2023
#> chen2022  2022          Scientometrics 10.1234/example.2022
```

1.4.3 Parsing WoS plain text with bibliometrix

```r
# Requires a WoS export file - not bundled
# wos_df <- convert2df("savedrecs.txt", dbsource = "wos", format = "plaintext")
# glimpse(wos_df)
```

1.4.4 Standardizing field names

```r
standardize_bib <- function(df) {
  df |>
    transmute(
      title = TITLE,
      authors = AUTHOR,
      year = as.integer(YEAR),
      journal = JOURNAL,
      doi = DOI
    )
}

bib_df |> standardize_bib()
```

```
#> # A tibble: 2 x 5
#>   title                                 authors    year journal            doi
#>   <chr>                                 <list>    <int> <chr>              <chr>
#> 1 Advances in Bibliometric Methods      <chr [2]>  2023 Journal of Inform~ 10.1~
#> 2 Citation Networks in Computer Science <chr [2]>  2022 Scientometrics     10.1~
```

1.4.5 Visualization

```r
completeness <- bib_df |>
  summarise(across(everything(), ~ mean(!is.na(.)))) |>
  pivot_longer(everything(), names_to = "field", values_to = "present") |>
  filter(present > 0) |>
  mutate(field = fct_reorder(field, present))

ggplot(completeness, aes(x = present, y = field)) +
```

```r
  geom_col(fill = palette_sci(1)) +
  scale_x_continuous(labels = scales::percent) +
  labs(x = "Completeness", y = NULL) +
  theme_sci()
```

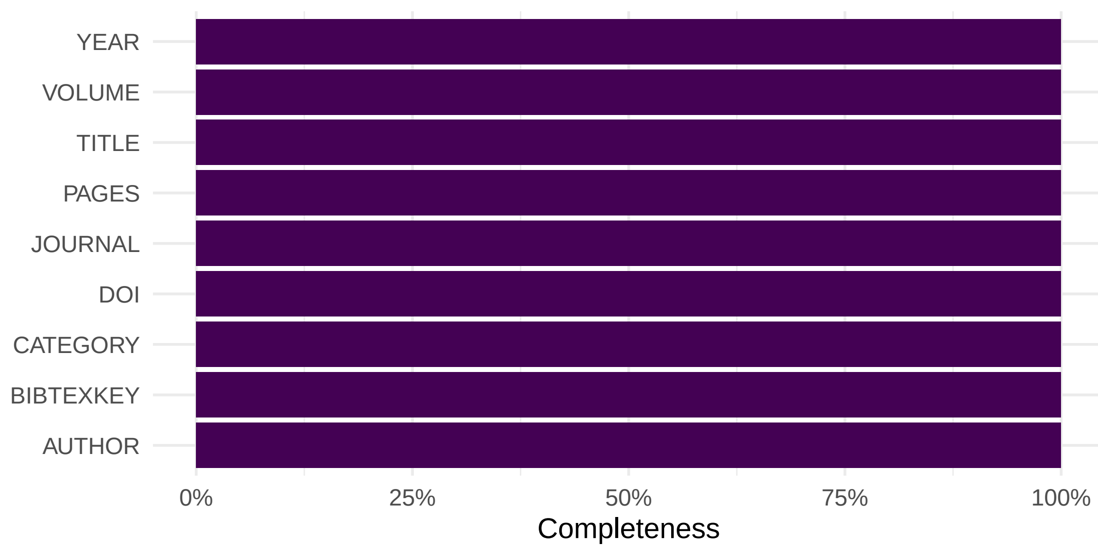

Figure 1: Field completeness in the parsed BibTeX records.

1.5 Diagnostics and interpretation

After parsing, always check:

- **Row count**: Does the number of records match the export? Missing entries may indicate parsing errors.
- **Field completeness**: Which fields are populated? Abstracts and keywords are often missing in BibTeX exports.
- **Author format**: Names may appear as "Last, First" or "First Last" depending on the source. Standardize before analysis.
- **Encoding**: Non-ASCII characters (diacritics, CJK) can corrupt during parsing. Ensure UTF-8 encoding.

1.6 Limitations and responsible use

1.7 Limitations and responsible use

- Export files are snapshots: they reflect the database state at export time and become stale immediately.
- Field mapping between formats is imperfect. Some formats lack fields (RIS has no standard abstract tag across all databases).
- Parsing errors are silent — always validate record counts and spot-check metadata against the source [@hicks2015].

1.8 Common pitfalls

1.9 Common pitfalls

- **Encoding mismatches**: Exporting as Latin-1 and reading as UTF-8 (or vice versa) corrupts names and titles.
- **Truncated exports**: Many databases cap exports at 500 or 1,000 records per file. Large corpora require multiple export batches.
- **Inconsistent author delimiters**: BibTeX uses "and", RIS uses newlines, CSV uses semicolons. Parsing must handle each.
- **Assuming format from extension**: A `.txt` file may be WoS plain text or something else entirely. Check the first few lines.

1.10 Exercises

1. **Export and parse.** Export 50 records from Google Scholar as BibTeX. Parse them with `bib2df`. How many fields are populated?

2. **Format comparison.** Export the same set of records from Scopus as both CSV and RIS. Parse both and compare the fields available.

3. **Standardization function.** Write a function that takes a data frame from `convert2df()` and returns a tibble with columns: `doi`, `title`, `authors`, `year`, `journal`, `cited_by`.

1.11 Solutions

Solutions are provided in 1.11.

1.12 Further reading

- @aria2017 — The `bibliometrix` package, including `convert2df()` for multi-format parsing.
- @priem2022 — OpenAlex as an API-first alternative to file-based workflows.

1.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.12.0
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8
#>  [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8
#>  [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
```

```
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] bibliometrix_5.4.0 RefManageR_1.4.0   bib2df_1.1.2.0     rcrossref_1.2.1
#>  [5] gt_1.3.0           tidytext_0.4.3     glue_1.8.1         openalexR_3.0.1
#>  [9] lubridate_1.9.5    forcats_1.0.1      stringr_1.6.0      dplyr_1.2.1
#> [13] purrr_1.2.2        readr_2.2.0        tidyr_1.3.2        tibble_3.3.1
#> [17] ggplot2_4.0.3      tidyverse_2.0.0    bookdown_0.46      fs_2.1.0
#> [21] rmarkdown_2.31
#>
#> loaded via a namespace (and not attached):
#>  [1] gridExtra_2.3          httr2_1.2.2            readxl_1.4.5
#>  [4] rlang_1.2.0            magrittr_2.0.5         otel_0.2.0
#>  [7] compiler_4.4.1         vctrs_0.7.3            httpcode_0.3.0
#> [10] pkgconfig_2.0.3        fastmap_1.2.0          backports_1.5.1
#> [13] labeling_0.4.3         ca_0.71.1              utf8_1.2.6
#> [16] promises_1.5.0         tzdb_0.5.0             tinytex_0.59
#> [19] xfun_0.57              jsonlite_2.0.0         SnowballC_0.7.1
#> [22] later_1.4.8            parallel_4.4.1         stopwords_2.3
#> [25] R6_2.6.1               stringi_1.8.7          RColorBrewer_1.1-3
#> [28] cellranger_1.1.0       Rcpp_1.1.1-1.1         knitr_1.51
#> [31] triebeard_0.4.1        base64enc_0.1-6        rentrez_1.2.4
#> [34] httpuv_1.6.17          Matrix_1.7-0           igraph_2.3.2
#> [37] timechange_0.4.0       tidyselect_1.2.1       stringdist_0.9.17
#> [40] dichromat_2.0-0.1      pubmedR_1.0.2          yaml_2.3.12
#> [43] viridis_0.6.5          codetools_0.2-20       miniUI_0.1.2
#> [46] humaniformat_0.6.0     curl_7.1.0             qpdf_1.4.1
#> [49] lattice_0.22-6         plyr_1.8.9             shiny_1.13.0
#> [52] withr_3.0.2            S7_0.2.2               askpass_1.2.1
#> [55] evaluate_1.0.5         zip_2.3.3              xml2_1.5.2
#> [58] shinycssloaders_1.1.0  pillar_1.11.1          janeaustenr_1.0.0
#> [61] DT_0.34.0              plotly_4.12.0          generics_0.1.4
#> [64] rprojroot_2.1.1        hms_1.1.4              scales_1.4.0
#> [67] xtable_1.8-8           contentanalysis_1.0.0  lazyeval_0.2.3
#> [70] tools_4.4.1            data.table_1.18.4      tokenizers_0.3.0
#> [73] openxlsx_4.2.8.1       pdftools_3.9.0         XML_3.99-0.23
#> [76] visNetwork_2.1.4       grid_4.4.1             bibtex_0.5.2
#> [79] urltools_1.7.3.1       rscopus_0.9.0          dimensionsR_0.0.3
#> [82] bibliometrixData_0.3.0 cli_3.6.6              rappdirs_0.3.4
#> [85] viridisLite_0.4.3      gtable_0.3.6           digest_0.6.39
#> [88] ggrepel_0.9.8          crul_1.6.0             htmlwidgets_1.6.4
#> [91] farver_2.1.2           htmltools_0.5.9        lifecycle_1.0.5
#> [94] httr_1.4.8             here_1.0.2             mime_0.13
```